

Design of Composite Radomes for Low-Altitude Wildfire Response UAVs

By Evan Mullins

EMA 3011 Design Project

1. Introduction

In 2024, 64,897 reported wildfires accounted for 8,924,884 acres worth of damage to American land. Of this total, 147 reported requesting assistance from unmanned aerial systems with a fulfillment rate of 61.2%. [1]. Though natural wildfires may appear beneficial in controlled amounts, unpredictable wildfires have created a unique threat to the physical and economic livelihood of those affected. These chaotic fires impede traffic routes, drain natural resources, disrupt strictly timed schedules, and threaten the lives of those both evacuated and called to respond.

As the dangers associated with wildfire relief grow, efforts turn to adapting modern technologies to revolutionize firefighting by improving efficacy and personal safety. One underutilized method of multiplying firefighter impact is the use of unmanned aerial vehicles (UAVs). These drones may be equipped with cameras (LiDAR or thermal) that send data for machine learning risk evaluations to identify fire trends and promote strong preventative and counter actions.



Figure 1. News coverage of UAS Wildfire Response: The California National Guard utilizes MQ-9s to analyze wildfires alongside human intervention. [2]

While precedent exists for the use of sophisticated UAVs such as General Atomics' MQ-9 "Reaper" to fight such fires, operating costs and proprietary information associated with activating bodies like the California National Guard raise concerns about response time and availability.

These challenges in adapting aerial response to the conditions of wildfire operation suggest a need for redesign of UAV components to sustain operational efficacy and recoverability in extreme environments. When deployed, drone operation is common above or around thick plumes to avoid particulate damage, external stresses, and thermal damage to internal components. Oftentimes, this region is selected for aerial clarity and durability of current composite units relying on polyurethane foam cores [3] that cannot withstand the high operating temperature of wildfires. One region of concern is the nosecone unit called a "radome," which houses internal electronics and ensures robust antenna signal. When adapting UAVs to withstand the erosion of plumed ashes and thermal fatigue of wildfire, engineers must emphasize structurally sound and thermally resistant materials to protect the electronics critical for sensing, data handling, and flight operation.

Ultimately, when redesigning a UAV for wildfire operation to maximize data collection, moving operating limits closer to the wildfire tree line may allow for entrance to a region fruitful for analysis and advanced sensing technologies that are often bottlenecked by height and survival. Consequently, this drive to fly lower demonstrates the need to select lightweight, RF-transparent materials with high service temperatures that can maintain a clear signal and structural integrity at lowered, critically hot altitudes while surviving particulate matter and returning intact for later redeployment.

2.1 Selection Criteria

When selecting radome materials for wildfire operation, the extreme operating conditions reveal three primary property regions of interest: mechanical (stiffness and density), electrical (RF transparency), and thermal properties (maximum service temperature, coefficient of thermal expansion, and thermal conductivity). These properties can then be classified according to their constraint level as either design-limiting constraints or properties used for design optimization.

Design-limiting thermal properties reveal themselves first, since the radome must maintain structural integrity while operating under cyclical thermal stresses. Wildfires may burn more than 1000 °C [4]. Because drones operate at an altitude far from the core flame, other studies report a significantly lower 120 °C as the operating temperature threshold near wildfires [5]. The ideal material will push this current operating much higher to around 298 °C, an engineering boundary set by the flashpoint of wood and representative of the ignition spread temperature. This new temperature envelope of 120-298 °C establishes a hard yet conservative design boundary for selection; any material with a maximum service temperature significantly under the greater boundary cannot be considered operationally safe. Thermal expansion and thermal conductivity then dictate the severity of heat transfer and degradation of the radome over time and can be treated as design-optimizing properties that must be minimized.

Electronic properties then dictate the ability of a UAV to maintain a clear signal while also being able to collect external data, depending on the sensor type and geometry. The two properties in question are the dielectric constant of the material and

its loss tangent. The dielectric constant acts as a hard constraint, since radome materials require a dielectric constant lower than $\epsilon_r = 4$ [6] to minimize interference. The loss tangent then acts as an optimizing constraint, and once again, a minimal value is desired.

Mechanical properties dominate in nosecone optimization, as maximizing the strength-to-weight ratio dominates aerial vehicle design. An optimal material will maximize Young's Modulus (stiffness) while minimizing density. While fracture toughness should also be considered, optimization of this property comes with diminishing returns. Radome materials should overcome a lower threshold of brittleness such that they avoid substantial crack propagation from debris that may lead to eventual catastrophic failure during flight.

Rank	Property
1 (Design-Limiting)	Maximum Service Temperature ($T_{\max}$)
1 (Design-Limiting)	Dielectric Constant (ϵ_r)
2 (Priority Optimizing)	Density (ρ)
3 (Priority Optimizing)	Young's Modulus (E)
4 (Priority Optimizing)	Coefficient of Thermal Expansion - CTE (α)
5 (Optimizing)	Loss Tangent ($\tan(\delta)$)
6 (Optimizing)	Thermal Conductivity (k)
7 (Survivability Threshold)	Fracture Toughness (K_{Ic})

Figure 2. Ranked Material Properties: A ranked list of the merit index properties based on the above justifications.

A few adjustments and clarifications are required before assembling a complete merit figure.

First, while the screening equation will consider the maximum service temperature, its primary function will be to enforce the lower limit of 298 °C. While this temperature is representative of the flashpoint of wood and conservatively higher than the 120 °C operating temperature of UAVs, designing a radome to withstand both improves altitude flexibility and safety margin. Due to uncertainty in recorded temperatures, materials that can surpass both ranges will be considered for merit, but materials over the 298 °C cutoff can be considered for an evolutionary decrease in operational altitudes.

Second, while screening for radome materials that operate with a presumed dielectric constant below $\epsilon_r = 4$, many carbon-fiber reinforced composites appear as likely candidates. However, due to the conductivity of carbon fiber [7], these reinforcement fibers cause these composite structures to disrupt and produce a lossy signal that makes them far from ideal for radome operation. While carbon fibers may appear ideal for the airframes or other structural components, they will be discarded in material screening. Finally, during preliminary screening, observation of the baseline operational values of polyurethane (PU) foam properties used in various aerospace grade radomes [3] assists in quantification of material baselines.

Polyurethane foam (rigid, closed cell, 0.062)				
Datasheet view:	All attributes	Show/Hide	Find Similar	
Thermal properties				
Maximum service temperature	i	135	- 177	°C
Minimum service temperature	i	-205	- -185	°C

Figure 3. Sample Identification of PU properties: This PU foam exhibits an average T_{max} of 156 °C. This data and subsequent data regarding material properties were obtained using Ansys Granta Edupack [8]

Because this baseline value is drawn from the maximum service temperature of a high-grade rigid PU foam, this preliminary value demonstrates the limits of foam use during wildfire operations. While average service temperatures fall mildly above the comfortable operating range, spikes in heat and an inability to withstand elevated temperatures prevent drones with PU from lowering their operation height without experiencing significant structural degradation.

Finally, the operating environment of a wildfire adds additional constraints to material selection that align more with geometry than material properties. Due to the small form factor required when navigating chaotic wildfire debris, large operation times, and the long distance of surveying, a wildfire relief drone should be built as a fixed wing, vertical take-off and landing (VTOL) vehicle akin to a small scale MQ-9. Second, with the likelihood of being damaged regardless of material strength, the design of small, replaceable nosecones dominates a VTOL-style redesign. Finally, binding materials like epoxy resins may be subject to uneven and unexpected degradation in such a high-temperature environment. As a result, screening preference will be given to materials that can be designed as quick, reproducibly manufactured monolithic nosecones instead of those contingent on lengthy, hybrid composite or sandwich style layups. Future discussion of sandwich radome designs that may significantly improve operating temperatures may be warranted but is beyond the scope of this selection.

2.2 Material Identification

With this baseline and the prior limits established, a preliminary evaluation of potential candidates can be performed by screening the two hard constraints against each other.

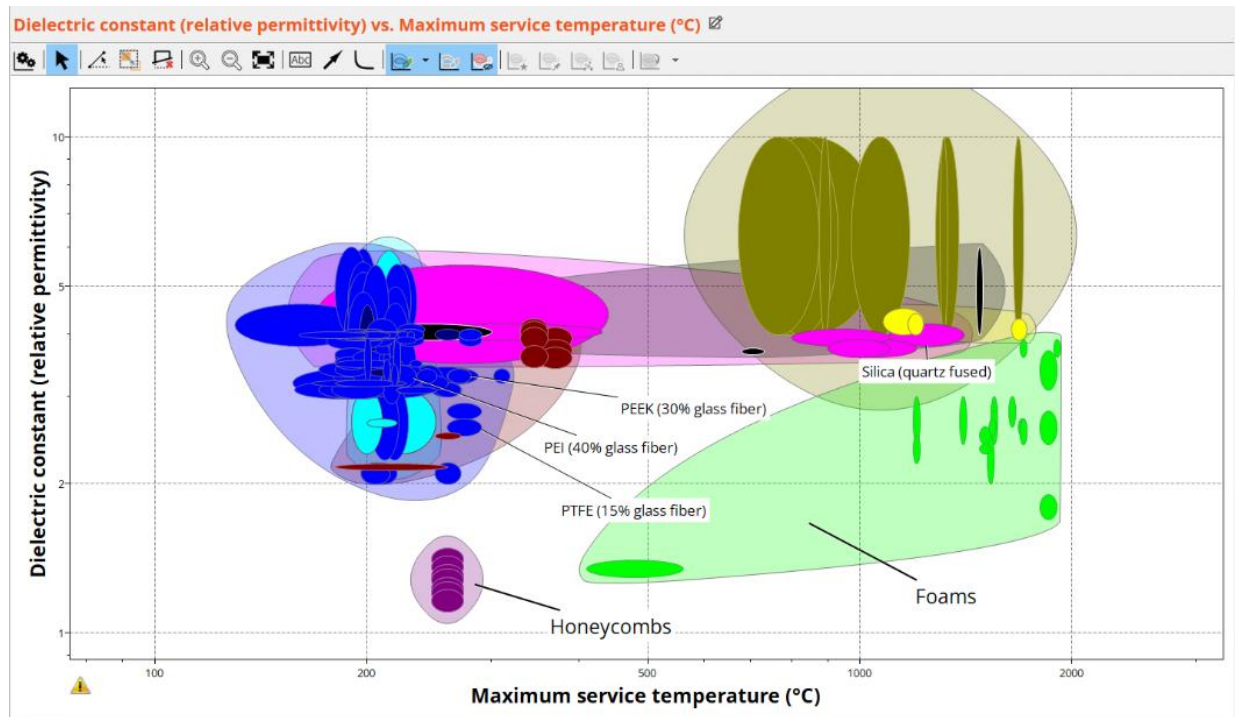


Figure 4. Preliminary Material Evaluation of Dielectrics vs Service Temperature: The screening hides records that fail design-limiting constraints, and it highlights composite plastics, honeycombs, foams, and glasses as candidates.

Despite the large list of candidates that appear, honeycombs and foams can be eliminated due to their purpose as the core of sandwich style layups instead of single-material nosecones. PTFE, while glass reinforced, has a low glass transition temperature of -103 °C with a significantly higher (yet still insufficient) second phase transition at 116 °C [9]. Granta Edupack notes that PTFE cannot be molded injection and has a relatively low Elastic Modulus (1.01 GPA vs 9.81 GPA for PEEK and 12.1 GPA for PEI), which makes use of this polymer blend difficult for manufacturing desired nosecone shapes. Finally, all PCB laminate materials consisting primarily of PBI are ignored due to an improper use case and their inability to be injection molded. As a result, the primary candidates that survive the initial screening are GF-PEI (40% glass fiber), GF-PEEK (30% glass fiber), and Silica (quartz fused).

After this primary screening, construction of a figure of merit combining selection criteria such that minimized properties use inverse relationships and maximized properties use direct relationships proves useful. Explicitly, Young's modulus and maximum service temperature should be maximized while density, CTE, thermal conductivity, and loss tangent should be minimized.

However, a few properties must be noted. When treating the nosecone as a bending panel, stiffness is proportional to Young's modulus and the cube of panel thickness. Because panel mass per area is proportional to thickness and density, substitution shows that the required thickness scales as $E^{-1/3}$ power. Consequently, to reduce thickness and mass, merit is based on $E^{1/3}$ rather than E . A complete proof of this exponent can be found in the appendix. Sensibility of selected materials can be verified using a typical drone density threshold between 1 and 2 g/cm³ [10]. Finally, rather than including fracture toughness in the merit equation, *Merit* will be plotted against an x-axis of toughness to separate materials that can survive harsh plumes, stresses, and debris. With these modifications in place, the merit index can be calculated using property averages. The equation and resulting screen appear as:

$$M = \frac{T_{max} \cdot E^{\frac{1}{3}}}{\tan(\delta) \cdot k \cdot \rho \cdot \alpha \cdot \epsilon_r}$$

Figure 5: Merit Index Equation: This equation is the multiplicative index used to rank materials based on the outlined design constraints. Its units are Pa^{1/3} m⁴ K³ W⁻¹ kg⁻¹.

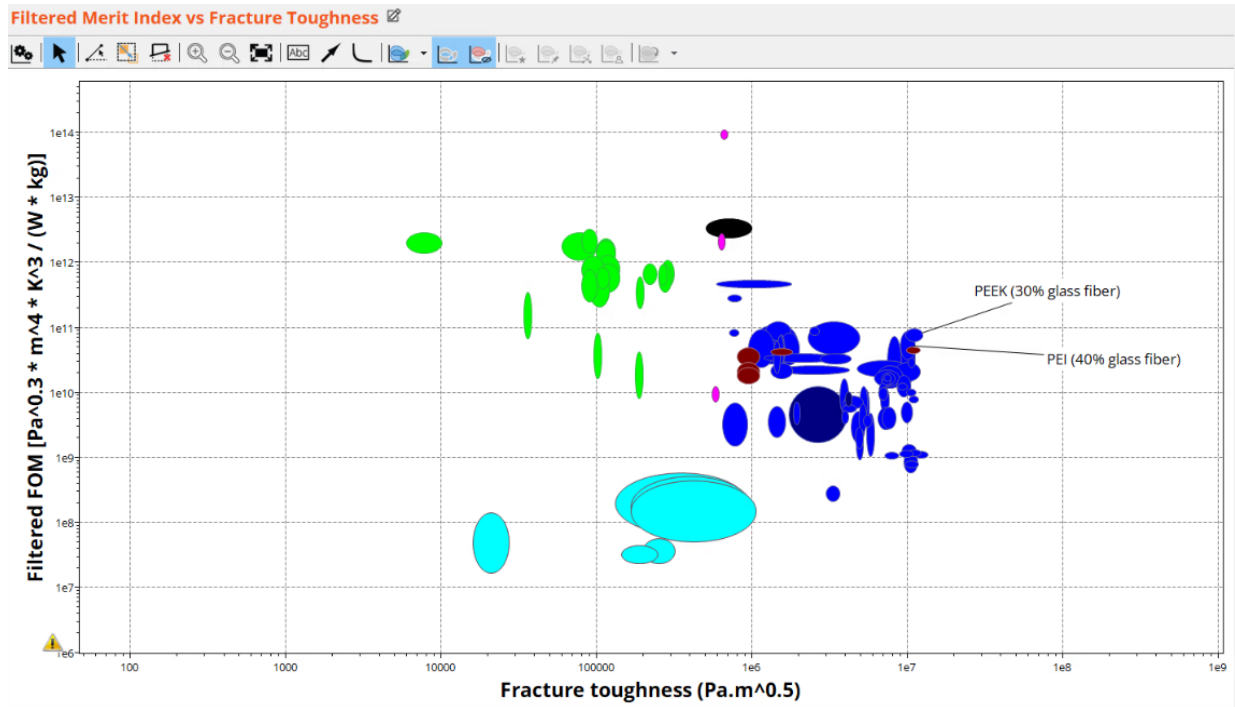


Figure 6. Final FOM vs Fracture Toughness: Glass families (pink) fall short of 1 MPa, while PEI and PEEK lead the merit index (top right). Because of the minimal differences between these candidates, calculation is required.

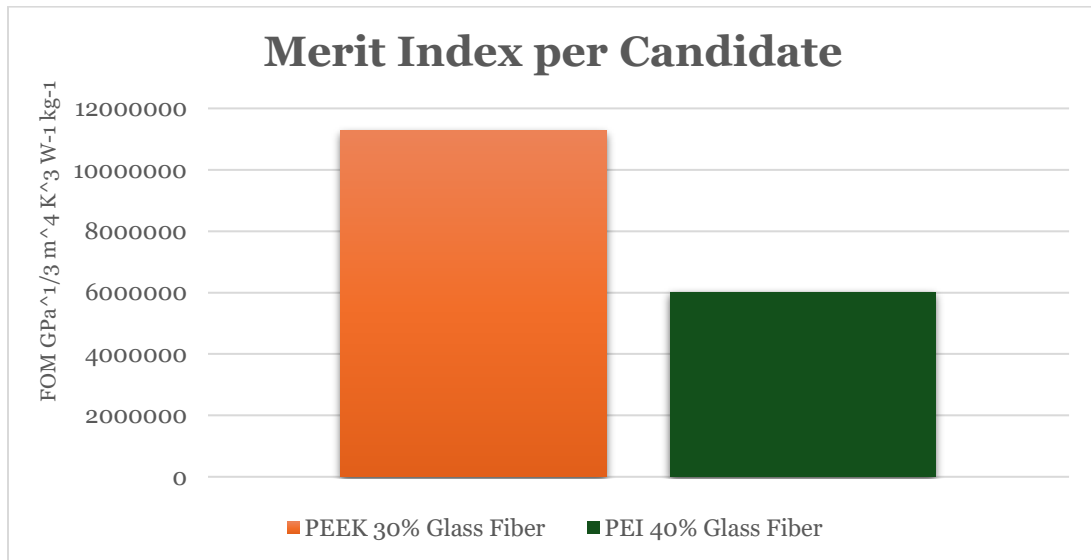


Figure 7. Merit Indices: PEEK appears of greater merit than PEI ($M = 1.13 \cdot 10^7 > 6.01 \cdot 10^6 \text{ GPa}^{1/3} \text{ m}^4 \text{ K}^3 \text{ W}^{-1} \text{ kg}^{-1}$). Values provided using GPa instead of Pa to reduce the order of each index. Comprehensive data in the Appendix.

Based on the figures above, glass families can be eliminated based on merit. This justification also comes intuitively, since, despite their superior thermal resistance and dielectric behavior, glasses tend to fail from brittleness due to crack propagation. This selection reveals two candidates: 30% GF-PEEK and 40% GF-PEI.

2.3 Recommendation and Discussion

Based on the merit index comparison, the physics-based property constraints, the geometric constraints, and the desired operating behavior of a radome material within a wildfire environment, 30% GF-PEEK appears to be a sufficiently justified candidate. This blend of PEEK falls within the common density range of traditional UAV materials with an average density of 1.52 g/cm³. Furthermore, this blend of PEEK can be injection molded, allowing for the manufacturing of monolithic radomes without the geometric difficulty associated with small-scale sandwich models.

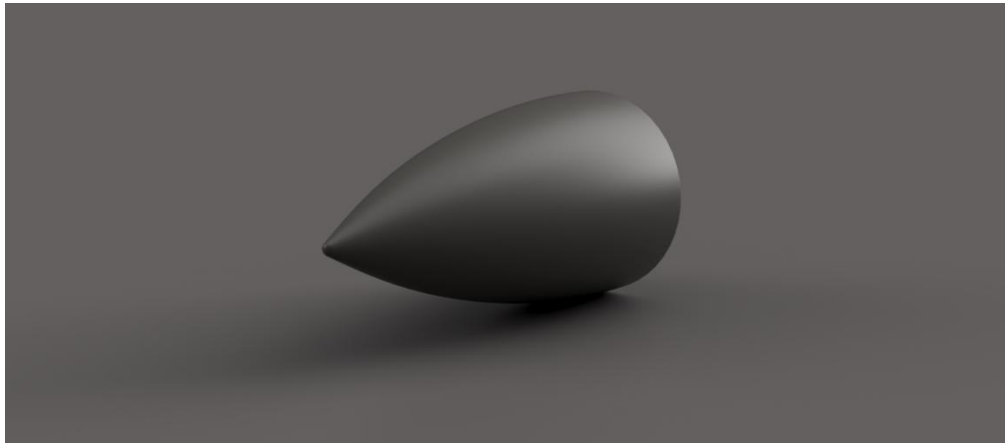


Figure 8. Fusion360 Rendering of a Blunted-Ogive-PEEK Nosecone. The grey coloring reflects the 30% fiberglass reinforcement throughout the radome.

However, PEEK comes with implementation challenges. While this material dominates the merit values, its maximum service temperature ranges from 260-280 °C with a melting temperature range of 322-346 °C. Depending on the actual operating

environment, this range of structural integrity could prove substantially overengineered or sufficiently underengineered. As a result, internal cooling systems, heat sinks, or heat shielding coatings may be necessary for UAV implementation, depending on the environment. One potential solution may utilize silica aerogels as a source of thermal insulation, but evaluation of the dielectric behavior, thermal efficiency, and material compatibility requires further inquiry beyond the scope of this study. Second, while PEEK has a lower dielectric constant ($\epsilon_r = 3.3$ vs 3.55) and a much higher average $T_{\max}$ (270 °C vs 220 °C), glass fiber PEEK is ultimately an extremely expensive material, costing roughly four times as much as glass fiber PEI (\$55/kg vs \$11/kg).

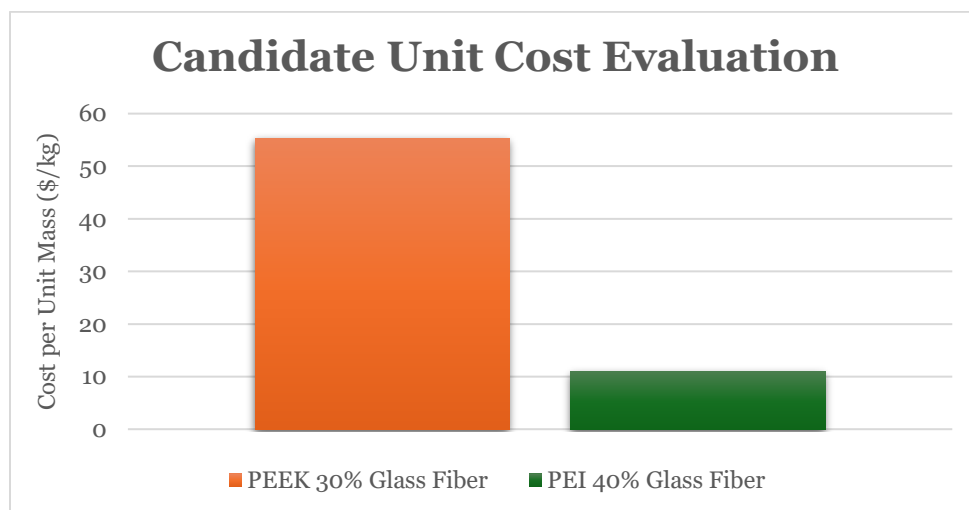


Figure 9. Cost/Unit Mass Breakdown: 30% GF- PEEK appears significantly more costly than PEI 40% GF by 4x.

Ultimately, while it may require iterative research and optimization, 30% GF-PEEK is sufficiently prepared for wildfires depending on the operating conditions. However, investment in an optimized radome material may prove a prudent long-term decision when considering the threat brought by wildfires and the lives at stake.

3. Conclusion

As wildfire response remains unmet by modern technology, research interests have intensified around adapting autonomous aerial systems to combat, contain and prevent spontaneous fires. Initiatives such as the \$11 million XPrize Wildfire Challenge accelerate interest in these relief platforms. Concurrently, academic efforts to improve UAVs for wildfire relief demonstrate a shift in attention towards this challenge. For example, Dr. Ifju and Dr. Bartos pioneered the University of Florida's XPrize "SAFE drone" that used UAVs to suspend fire extinguishers. Labs augmenting UAVs with Artificial Intelligence and researching the deployment of swarms specifically for wildfire relief demonstrate the versatility of responding to this problem. [11-13]

This study demonstrates how adapting a fixed-wing VTOL design for low-altitude operation becomes possible when redesigning the radome unit around high-temperature, RF-compatible composite materials. By selecting a glass-reinforced polymer such as 30% GF-PEEK, the radome unit can protect onboard electronics and RF systems at lower altitudes with temperatures over 100 °C greater than traditional polyurethane foam designs. Additionally, the injection moldability of this PEEK blend allows for rapid manufacturing of radomes for fleet-based missions. Finally, these radome units can be redeployed in standard operating conditions to maintain similar efficiency, allowing them to become a versatile platform for UAV operation.

While adoption of this technology may appear slow and costly, the long-term benefits of understanding wildfire intelligence are substantial. Advancing UAV capabilities for wildfire prevention will mitigate environmental loss, economic damage, and, most importantly, the cost of lives lost by wildfire victims and responders.

References

- [1] National Interagency Coordination Center, “National Interagency Coordination Center Wildland Fire Summary and Statistics Annual Report 2024,” 2024. [Online]. Available: https://www.nifc.gov/sites/default/files/NICC/2-Predictive%20Services/Intelligence/Annual%20Reports/2024/annual_report_2024.pdf. [Accessed: Apr. 21, 2026].
- [2] General Atomics Aeronautical Systems, “MQ-9 Mission Range Across Government: From Security to Wildfire,” Image 3: (KCRA 3 2018 news broadcast), 2024. [Online]. Available: <https://www.ga-asi.com/mq-9-mission-range-across-government-from-security-to-wildfire>. [Accessed: Apr. 21, 2026].
- [3] J. Meyer “Polyurethane Foam: Dielectric Materials for Use in Radomes and Other Applications,” General Plastics Manufacturing Company, White Paper, 3 Nov. 2015. [Online]. Available: https://www.generalplastics.com/wp-content/uploads/2016/11/White-Paper-PU-Foam-Dielectric-Materials-for-Use-in-Radomes-and-Other_Applications.pdf. [Accessed: Apr. 21, 2026]
- [4] U.S. Geological Survey, “How Hot Are Wildfires?” 2024. [Online]. Available: <https://www.usgs.gov/media/images/how-hot-are-wildfires>. [Accessed: Apr. 21, 2026].
- [5] S. Conley *et al.*, “VTOL Analysis for Emergency Response Applications (VAERA) – Identifying Technology Gaps for Wildfire Relief Rotorcraft Missions,” NASA Ames Research Center, Moffett Field, CA, USA, Tech. Report 1682, 23 Jan. 2024. [Online]. Available: https://rotorcraft.arc.nasa.gov/Publications/files/1682_Conley_Final_012324.pdf. [Accessed: Apr. 21, 2026.]
- [6] N. S. Pendergrass, “Radome Analysis,” Technical Report SD83-BMDSCOM-2682, Department of the Army, Ballistic Missile Defense Command, Systems Technology. Project Office, Huntsville, AL, USA, Oct. 1983. [Online]. Available: <https://apps.dtic.mil/sti/citations/ADA136805>. [Accessed: Apr. 21, 2026].
- [7] I. M. Alarifi, “Investigation the conductivity of carbon fiber composites focusing on measurement techniques under dynamic and static loads,” *Journal of Materials Research and Technology*, vol. 8, no. 5, pp. 4863–4893, 2019, doi: 10.1016/j.jmrt.2019.08.019.
- [8] Granta Design Limited, “Granta EduPack 2024,” Ansys, Cambridge, UK, 2024.
- [9] G. Calleja, A. Jourdan, B. Ameduri, and J.-P. Habas, “Where is the glass transition temperature of poly(tetrafluoroethylene)? A new approach by dynamic rheometry and mechanical tests,” *European Polymer Journal*, vol. 49, no. 8, pp. 2214–2222, 11 May 2013, doi: 10.1016/j.eurpolymj.2013.04.028.
- [10] A. Kantaros, C. Drosos, M. Papoutsidakis, E. Pallis, and T. Ganetsos, “Composite filament materials for 3D-printed drone parts: Advancements in mechanical strength, weight optimization and embedded electronics,” *Materials*, vol. 18, no. 11, p. 2465, 24 May 2025, doi: 10.3390/ma18112465.
- [11] University of Florida, “XPRIIZE at UF,” 2024. [Online]. Available: <https://explore.research.ufl.edu/xprize-uf.html>. [Accessed: Apr. 21, 2026].
- [12] S. P. H. Boroujeni *et al.*, “A comprehensive survey of research towards AI-enabled unmanned aerial systems in pre-, active-, and post-wildfire management,” *Information Fusion*, vol. 108, p. 102369, 2024, doi: 10.1016/j.inffus.2024.102369.
- [13] E. Ausonio, P. Bagnerini, and M. Ghio, “Drone swarms in fire suppression activities: A conceptual framework,” *Drones*, vol. 5, no. 1, p. 17, 2021, doi: 10.3390/drones5010017.

Appendix

The objective function for the mass of the panel is the same as that for the tie:

$$m = AL\rho = bbL\rho \quad (5.18)$$

Its bending stiffness, S , is given by equation (5.5). It must be at least:

$$S^* = \frac{C_1 EI}{L^3} \quad (5.19)$$

The second moment of area, I , for a rectangular section (Figure 5.2) is

$$I = \frac{bb^3}{12} \quad (5.20)$$

The stiffness S^* , the length L and the width b are specified; only the thickness b is free. We can reduce the mass by reducing b , but only provided the stiffness constraint is still met. Using the last two equations to eliminate b in the objective function for the mass gives

$$m = \left(\frac{12S^*}{C_1 b} \right)^{1/3} (bL^2) \left(\frac{\rho}{E^{1/3}} \right) \quad (5.21)$$

The quantities S^* , L , b and C_1 are all specified; the only freedom of choice left is that of the material. The best materials for a light, stiff panel are those with the smallest values of $\rho/E^{1/3}$ (again, so long as they meet any other constraints). As before, we will invert this, seeking instead large values of the material index M_p for the panel:

$$M_p = \frac{E^{1/3}}{\rho} \quad (5.22)$$

Figure 10. Derivation of Stiffness Index for Panels (Ashby p. 111): Used to evaluate the stiffness of the nosecone when treated as a panel.

Candidates	Young's Modulus (Gpa)	Maximum Service Temperature (C)
PEEK 30% Glass Fiber	9.81	270
PEI 40% Glass Fiber	12.4	224

Density (kg/m^3)	CTE (strain/C)	Thermal Conductivity
1515	0.000025	0.205
1585	0.00001505	0.295

\$/kg	Dielectric Constant	Loss Tangent
55.35	3.3	0.002
11.05	3.55	0.00345

Figure 11. Table of Properties: Calculated by averaging high and low data from Ansys Granta, then plugged into FOM (neglecting \$/kg).